

TYPES OF TANNERY EFFLUENT

Leather tanning process

Tanning is the process which converts the protein of the raw hide or skin into a stable material, which will not putrefy and is suitable for a wide variety of end applications. The two main types of tanning are chrome tanning and vegetable tanning. The highly polluting chromium is the most commonly used tanning material producing leather that is more supple and pliable than vegetable-tanned leather, and does not discolor or lose shape in water as drastically as vegetable-tan. Chromium compounds are applied to protect hides and skins from decay and to make them more durable against moisture and aging. Chromium interacts with fibers in the raw hide/skin during a bathing process, after which the tanned hides are wrung and prepared for finishing.

Major process sequences and chemical inputs

Curing of hides and skins

In most developed countries hides and skins are removed and lightly cured with salt at abattoirs or local hide collection centers. Preservation by chilling is also becoming more common in many places. Throughput is relatively regular, and technical supervision is available. The hides and skins then enter commercial channels and are transported to tanneries for processing.

In developing countries, it is also common for animal slaughter to take place at rural village sites. The rate of production of hides and skins in these countries is usually low and erratic, as it is governed partly by seasonal conditions. Under such circumstances, and with high ambient temperatures, curing is not simple. Salt, if available, is applied and the skins are then dried (dry salting). A major drawback of drying is that the tanner must subsequently reverse the process (rehydrate). Soaking the skins requires large volumes of water. In some countries salt curing is carried out by "wet salting", whereby the hide is immersed and saturated with brine solution.

Curing often incorporates treatment with insecticides to discourage beetle and other insect attack during storage and transport. The later washout of these chemicals may cause problems with effluent quality, as few wastewater treatment systems are designed to remove such substances.

Insecticides may include: pyrethrum, permethrin, p-dichloro-benzene, sodium silico-fluoride, borax.

Earlier insecticides such as DDT, benzene hexachloride and dieldrin are banned for such use in many industrialized countries because of their combination of persistence and environmental toxicity. Arsenic and mercury insecticides should not be used anywhere on account of their high human and environmental toxicity, and their extreme persistence.

Soaking

The first tannery operation in beamhouse where raw hides and skins are treated with water to rehydrate, clean, and soften them for subsequent processing. Wet-salted hides and skins, lose nearly 30-35% of natural water by curing. Soaking for few hours in plain water will rehydrate wet salted hides and skins and bring the stock to a condition suitable for the next operations.

The main chemicals used for soaking purpose are water, wetting agents (usually anionic or nonionic surfactants), soda ash (sodium carbonate (Na_2CO_3)), bactericides, enzymes (protease base) etc.

Unhairing and Liming

Unhairing/dehairing is the process to remove hair by means of chemical or biochemical treatments. Generally, unhairing includes wool/bristle. Liming is a general term for an alkaline treatment that causes swelling and hydrolysis of the pelt, but should not be excessive. Additionally, the process opens the fibre structure of hides. The duration of the process may vary from 18 hours to few days. The process is responsible for the major part of the COD load from a tannery.

The main chemicals used for liming purpose are lime, sodium sulfide, sodium hydrosulfide, liming auxiliary, liming enzymes, degreasing/wetting agent etc.

Fleshing

Fleshing is adipose tissue, which is loose connective tissue dominated by fat cells, or adiposities, and is removed either manually or mechanically from plumped hide or skin after the liming process. The fleshing operation results in a clean, uniform surface. The flesh removed from the hide prior to unhairing and liming, particularly if the hides have a large amount of tissue is called green fleshing. Lime fleshing which is carried out after liming.

Deliming

To remove lime partially or fully from the pelts is known as deliming. The pelt contains lime at 0.5-2.0% by weight. It is clear that the reaction involves lowering the pH to neutralize residual alkali from the pelts. The objectives of deliming is to lower the pH to 8.5-9.0, in preparation for bating, remove of free and chemically combined lime, reduce the swelling of pelts to the expected degree etc.

The main chemicals used for deliming process are ammonium salt, weak acid, strong acid, carbon dioxide etc.

Bating

Treating hides with enzymes to soften them while maintaining their flexibility and firmness. In this process, generally, proteases are used to remove scud and unwanted proteins. Usually, the non-structural proteins are the target. These proteins can be degraded by general proteases, because

they do not have highly defined structure. The bating material are typically composed of wood floor or another carrier, deliming agent and pancreatic enzyme.

Pickling

The final beam house operation is pickling. The pickling process is primarily conducted to adjust the collagen to the conditions required by the chrome (or any other) tanning reaction. Here, delimed and /or bated pelt are treated with acid and salt (Acidification of the pelt).

The main chemicals used for pickling are float, common salt, organic acids, sulfuric acid, CO, hypo etc.

Tanning

Tanning is the chemical and physical process of converting raw animal hides into durable, supple, and rot-resistant leather. It permanently alters the protein structure (collagen) of the skin so that it no longer decays. The process typically involves pretanning, tanning, and finishing. Depending on the desired outcome, different methods are used:

- **Mineral/Chrome Tanning:** The most common commercial method. It uses chromium salts to penetrate the hide, creating highly flexible, water-resistant, and uniformly colored leather. This process is very fast (takes only hours).

After pickling, when the pH is low, chromium salts (Cr^{3+}) are added. To fixate the chromium, the pH is slowly increased through addition of a base. The process of chromium tanning is based on the cross-linkage of chromium ions with free carboxyl groups in the collagen. It makes the hide resistant to bacteria and high temperature. The chromium-tanned hide contains about 2-3 dry weight percent of Cr^{3+} . Wetblue, i.e. the raw hide after the chrome-tanning process, has about 40 percent of dry matter.

- **Vegetable Tanning:** An ancient, eco-friendly method that uses natural tannins extracted from tree bark (e.g., oak, chestnut). It creates firm, thick, classic leather that ages beautifully but can take weeks to months to complete.

Vegetable tannins are polyphenolic compounds of two types: hydrolysable tannins (i.e. chestnut and myrobalan) which are derivatives of pyrogallols and condensed tannins (i.e. hemlock and wattle) which are derivatives from catechol. Vegetable tanning probably results from hydrogen bonding of the tanning phenolic groups to the peptide bonds of the protein chains. In some cases as much as 50% by weight of tannin is incorporated into the hide

- **Oil/Brain Tanning:** Used historically for buckskin and heavy fur pelts. This process uses emulsified fats (such as animal brains, eggs, or neat's-foot oil) to coat the skin fibers, preventing them from sticking together as the hide dries, yielding a soft, breathable buckskin.

- **Synton:** Syntans are increasingly employed alone or in association with chrome and vegetable tannins, either for retanning or as principal tanning agents for certain speciality leathers. Syntans are typically sulphonated products of phenol, cresol and naphthalene, or resin derived from polyurethanes or polyacrylic acids.

Post tanning

Post-tanning is the critical series of chemical and mechanical processes that transform freshly tanned, rigid hides into soft, flexible, and aesthetically pleasing commercial crust leather. It involves neutralizing acidity, adjusting thickness, retanning for bulk, dyeing, oiling, and drying.

The post-tanning sequence follows a highly structured, step-by-step methodology:

1. Mechanical Preparation

- **Sammying:** The wet, newly tanned hide is mechanically squeezed using felt or rubber rollers to reduce moisture content to about 40–45%.
- **Splitting & Shaving:** The hides are fed through specialized industrial machines to shave and split them into a uniform, consistent thickness tailored to the end product (e.g., shoe uppers, garments, or upholstery).

2. Chemical Processing

All subsequent chemical treatments are typically done inside rotating drums based on the shaved weight of the hide.

- **Neutralization:** Tanned hides are naturally acidic. Weak bases (such as sodium bicarbonate) or syntans are used to neutralize this acid, balancing the leather so it can absorb dyes and fatliquoring agents.
- **Retanning:** Synthetic tannins (syntans), resins, or vegetable tannins are added to fill loose, empty parts of the hide structure, improving firmness and surface texture.
- **Dyeing:** Colorants are applied to achieve the desired aesthetic finish. Mainly acid, basic, direct or other dye are used
- **Fatliquoring:** Essential oils and fats are introduced to lubricate the individual fibers of the collagen matrix, preventing the leather from drying into a stiff board.

3. Drying and Softening

- **Setting Out & Drying:** The wet hides are mechanically smoothed to remove wrinkles and excess surface water, then dried (either naturally or via vacuum dryers).
- **Staking:** The dried, rigid leather is mechanically massaged (staked) to break apart the lubricated fibers, making it soft and supple.

Finishing

Leather finishing is the final stage of the leather-making process. It involves mechanical and chemical treatments to protect the hide, enhance its aesthetic appeal (color, gloss, texture), and tailor its tactile feel, durability, and performance for specific end-use applications. The most critical chemical groups include:

- Resins (Acrylic, Butadiene, Polyurethane): Form the foundational binder and protective film, providing toughness, flexibility, and a tight grain.
- Pigments: Opaque colorants used to coat the surface, hide natural defects, and provide a uniform color.
- Dyes: Translucent coloring agents that either alter the shade or are used in topcoats to maintain the leather's natural look.
- Nitrocellulose (NC) Lacquers: Applied as a transparent, water-resistant, and easily cleanable topcoat.
- Waxes and Oils: Used to adjust the final hand-feel and gloss of the leather, making it softer or enhancing its shine.
- Auxiliaries (Viscosity modifiers, defoamers): Help control the thickness and prevent bubbles during the application of the finishing mixture.

Pollution generation during several process

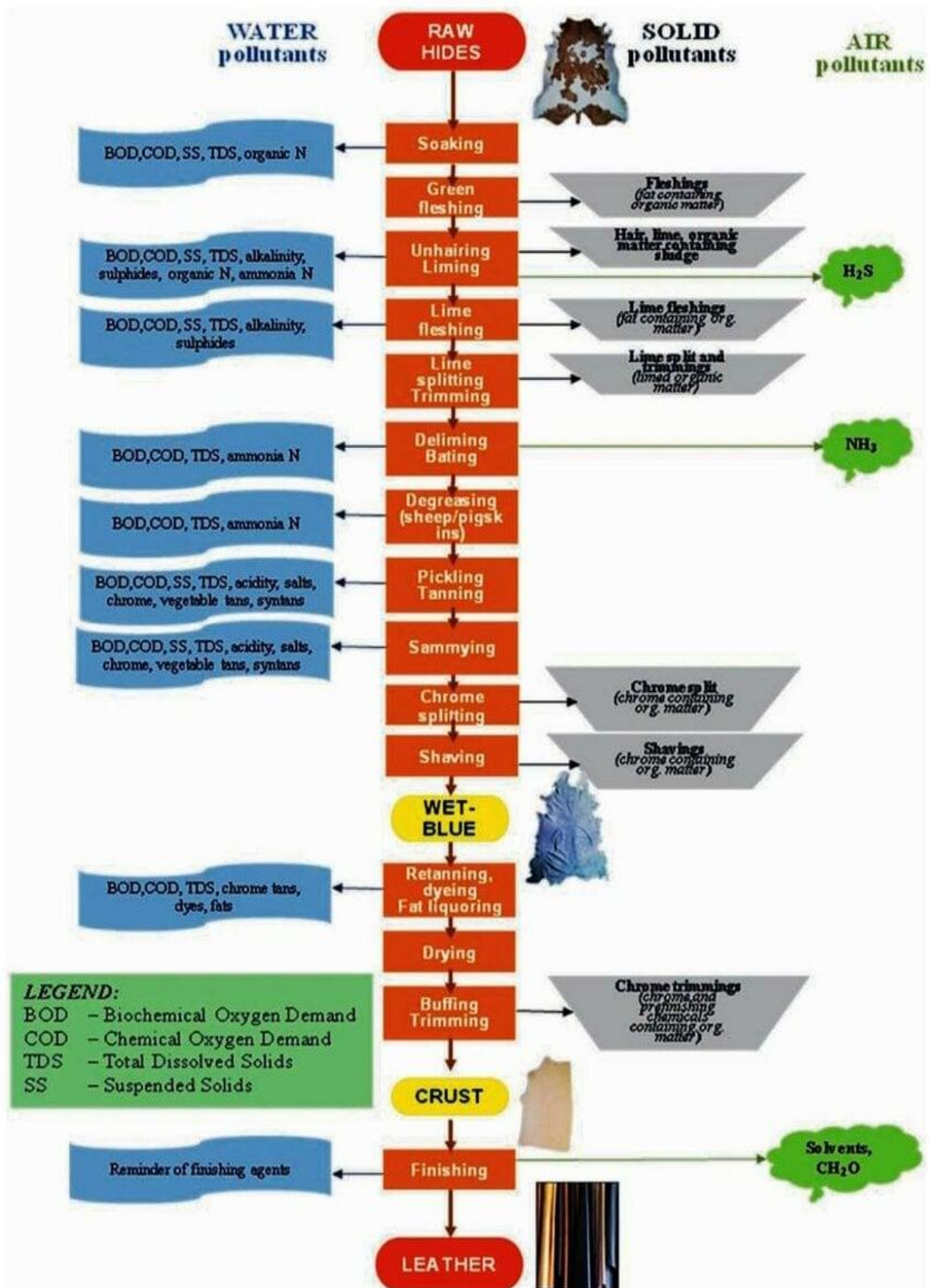
The following figure shows the pollution load generated from several process.

Beamhouse effluent—generated during hide soaking, unhairing, liming, deliming, and bating—accounts for up to 85% of a tannery's total pollution load. It is a highly toxic, alkaline wastewater containing extreme levels of suspended solids, sulfides, ammonia, and organic matter. Proper treatment is crucial to mitigate environmental and public health hazards.

Primary Constituents and Characteristics

The beamhouse process removes hair, natural fats, and non-collagenous proteins from raw hides. The resulting wastewater is defined by several key parameters:

- Sulfide and High pH: The unhairing/liming process uses sodium sulfide and lime, pushing the pH above 12. This creates toxic, odorous hydrogen sulfide gas and highly alkaline wastewater.
- Organic Load (BOD and COD): Dissolved hide proteins, fats, and residual hair lead to massive Chemical Oxygen Demand (COD) and Biological Oxygen Demand (BOD).



- **Ammonia and Nitrogen:** Deliming and bating operations use ammonium salts, resulting in high concentrations of ammonia-nitrogen (NH₃-N) and Total Kjeldahl Nitrogen (TKN).
- **Total Dissolved Solids (TDS):** Saline effluents from the soaking stage of wet-salted hides result in high salinity and chlorides

Tanyard (Tanning and Re-tanning): Produces acidic wastewater rich in heavy metals like trivalent chromium [Cr³⁺] and salts.

Post-tanning (Dyeing, Fatliquoring): Contributes synthetic dyes, oils, and organic solvents to the overall effluent stream

Emissions

1. Solid waste
2. Wastewater
3. Air pollution

The discharge of solid waste and wastewater containing chromium is the main environmental problem. Chromium is a highly toxic compound and the dumping of chromium containing material is in most countries restricted to a few special dumping grounds. Reduction of chromium discharge is therefore essential. Emissions into the air are primarily related to energy use, but also the use of organic solvents and dyes causes emissions into the air.

1. Solid waste

The production of fresh hides has been estimated at about 8-9 million tonnes per year (FAO, 1990a). During the processing of these hides a total of 1.4 million tonnes of solid waste is produced. This means that in all likelihood ca 16% of the processed hides is leather waste. Buljan (1994) puts the figures for trimmings and splittings (i.e. leather waste) at a total of 225 kg/ton hide (i.e. ca 23%). This is almost the same amount of waste produced as meat from fleshing activities (7 - 23%). For every ton of raw hide processed, the amounts of solid waste and by-products may be produced as given in Table 1. These figures show that the solid waste produced per ton of raw hide is about 450-600 kg. About half of this contains 3% chrome on a dry matter basis.

	Pre-tanning	Tanning	Finishing
Trimmings	120	110	32
Fleshings	70-230		
Wet blue split		115	
Buffing dust			2
Total	190-350	225	34
GRAND TOTAL	Approx. 450-600		

2. Wastewater

As for the production of wastewater, over 80 per cent of the organic pollution load in BOD terms emanates from the beamhouse (pretanning); much of this comes from degraded hide/skin and hair matter. The beamhouse is also the source of all non-limed and limed solid waste such as fleshing, trimming and waste split. As already mentioned, during the tanning process at least ca 300 kg of chemicals (lime, salt etc.) are added per ton of hides. Excess of non-used salts will appear in the wastewater. Because of the changing pH, these compounds can precipitate and contribute to the amount of solid waste or suspended solids.

Composition of tanner waste water.

The characteristics of tannery effluent vary considerably from tannery to tannery depending upon the size of the tannery, chemicals used for a specific process, amount of water used and type of final product produced by a tannery. The tannery waste contains large number of pathogens and foul smelling liquids containing both alkaline and acidic liquors

i. Solids

The solids to be found in tannery effluent fall into several distinct categories, solids are removed from tannery waste mechanically.

a. Suspended solids

The suspended solids component of an effluent is defined as the quantity of insoluble matter contained in the wastewater. These insoluble materials cause a variety of problems when discharged from a site; essentially, they are made up of solids with two different characteristics.

b. Solids with a rapid settling rate (settleable solids)

Settleable solids comprise material that can be seen in suspension when an effluent sample is shaken, but settles when the sample is left to stand. The majority of these solids settle within 5 to 10 minutes, although some fine solids require more than an hour to settle. These solids originate from all stages of leather making; they comprise fine leather particles, residues from various chemical discharges and reagents from different waste liquors. The settleable solids content is determined by leaving the shaken sample to settle and then filtering a known volume of the semi-colloidal matter remaining in suspension. After drying and weighing, the quantity of semi-colloidal matter can be calculated. The difference between the suspended solids and this figure is the settleable solids content.

c. Semi-colloidal solids

Semi-colloidal solids are very fine solids that, for all practical purposes, will not settle out from an effluent sample, even after being left to stand for a considerable period of time. They can, however, be filtered from solutions. Together with the more readily settleable solids, they thus comprise the suspended solids of an effluent that can be measured analytically. Most of these solids are protein

residues from the beam-house operations - mainly liming processes; however, large quantities are also produced owing to poor up take in vegetable tanning processes, another source being poor uptake during retanning.

d. Gross solids

Gross solids are larger than a sampling machine can handle, hence they are not measured. Their presence, however, is clear to see and the dangers they pose are fully recognized. The waste components that give rise to this problem are often large pieces of leather cuttings, trimmings and gross shavings, fleshing residues, solid hair debris and remnants of paper bags. They can be easily removed by means of coarse bar screens set in the waste water flow. If, however, they emerge from the factory, they settle out very rapidly.

ii. Chromium compounds

Two forms of chrome are associated with the tanning industry

a. Chrome 3(trivalent chrome, chrome III)

Chromium is mainly found in waste from the chrome tanning process. Despite the thermodynamic stability of Cr (III), the presence of certain naturally occurring minerals, especially MnO_2 oxides, can enhance oxidation of Cr (III) to Cr (VI). At high pH, Cr (VI) is bio available, and it is this form that is highly mobile and therefore poses the greatest risk of groundwater contamination.

b. Chrome 6+ (hexavalent chrome, chrome VI)

Tannery effluents are unlikely to contain chromium in this form. Chrome in the hexa-valent form is very toxic. Dichromates are toxic to fish life since they swiftly penetrate cell walls. Cr (VI) is a strong oxidant and therefore more reactive than Cr (III)

iii. Turbidity

Turbidity in water is caused by suspended and colloidal matter such as clay, silt, finely divided organic and inorganic matter, and plankton and other microscopic organisms. Turbidity is a measure of the clarity of a water body and is an optical measurement that compares the intensity of light scattered by a water sample with the intensity of light scattered by a standard reference suspension. It is commonly recorded in nephelometric turbidity units.

iv. Oxygen demand

Many components in effluents are broken down by bacterial action into more simple components. Oxygen is required for both the survival of these bacteria (aerobic bacteria) and the breakdown of the components. Depending on their composition, this breakdown can be quite rapid or may take a very long time. If effluent with a high oxygen demand is discharged directly into surface water, the sensitive balance maintained in the water becomes overloaded. Oxygen is stripped from the water causing oxygen dependent plants, bacteria, fish as well as the river or stream itself to die.

The outcome is an environment populated by non-oxygen dependent (anaerobic) bacteria leading to toxic water conditions. A healthy river can tolerate substances with low levels of oxygen demand. The load created by tanneries, however, is often excessive, and the effluent requires treatment prior to discharge. This is often achieved by using bacteria in a properly operated effluent treatment plant: a process demanding high levels of oxygen. Oxygen induction can be achieved by blowing large volumes of air into the effluent: a process entailing a high-energy demand and, as a corollary, high capital and operational costs. Under normal working conditions, both water and carbon dioxide are produced in large volumes; the process, however, depends upon bacterial growth. As the bacteria die, they form sludge that has to be treated and ultimately disposed of. This sludge has high water content and is often quite difficult to dewater, thus adding considerably to the treatment costs. In order to assess an effluent's impact on discharge to surface waters or determine the costs of treatment, the oxygen demand needs to be determined. This can be achieved in two different ways:

a. Biochemical oxygen demand (BOD₅)

The technique for measuring biochemical oxygen demand (BOD) is complex. Essentially, the effluent sample once shaken is left to stand for one hour so that all settleable solids are excluded from the analysis. The liquor above the precipitate (supernatant) is drawn off and used in the analysis. A suitable volume of this sample is diluted in water, pH adjusted, and seeded with bacteria (often settled sewage effluent). The samples are then incubated in the dark for five days at 20°C. The oxygen dissolved in the water is used by the bacteria while over time the organic matter in the sample is broken down. The oxygen remaining is determined either by means of an oxygen meter or by analysis. The level of oxygen demanded by the effluent can be calculated by comparison to the blank effluent-free samples.

The BOD₅ analysis, generally termed BOD, is widely used to assess the environmental demands of waste water. This method of detection has various weaknesses: the bacterial cultures can vary and the analysis is a highly sensitive process. If the most stringent care is not taken during the preparation and the analysis itself, the results can be misleading.

It should also be remembered that although BOD is a measure of the oxygen requirements of bacteria under controlled conditions, many effluent components take longer than the period of analysis to break down. Some chemicals will only be partially broken down, while others may not be significantly affected. Typically, vegetable tanning wastes have a long breakdown period, often quoted as being up to 20 days. These longer digestion periods can apply to a variety of the chemicals used in manufacturing leathers, including certain retanning agents, some synthetic fatliquors, dyes and residual proteins from hair solubilisation.

This longer breakdown period means that the environmental impact is spread over a larger area as the waste water components are carried greater distances before breaking down.

b. Chemical oxygen demand (COD)

This method measures the oxygen required to oxidise the effluent sample wholly. It sets a value for the materials that would normally be digested in the BOD₅ analysis, the longer-term biodegradable products, as well as the chemicals that remain unaffected by bacterial activity.

The method is very aggressive. A suitable volume of effluent is boiled with a powerful oxidizing agent (potassium dichromate) and sulphuric acid. As the effluent components oxidize, they use oxygen from the potassium dichromate, the amount used being determined by titration.

This method is often favored as it provides rapid results (hours as opposed to days). It is more reliable and cost effective as it is easier to manage larger numbers of samples.

The results are always higher than those obtained using the BOD₅ analysis. As a rule of thumb, the ratio between COD: BOD is 2.5:1, although in untreated effluent samples variations can be found as great as 2:1 and 3:1. This depends on the chemicals used in the different leather making processes and their rate of biodegradability.

It should be noted that both techniques are based on settled effluent, not filtered. The semi colloidal material that forms part of the suspended solids is also included in the BOD and COD determinations. Normally 1 mg/l suspended solids will generate a COD increase of approximately 1.5 mg/l.

v. Nitrogen

Nitrogen is contained in several different components in tannery effluent. Sometimes these sources have to be differentiated.

Total Kjeldahl nitrogen (TKN)

Several components in tannery effluent contain nitrogen as part of their chemical structure. The most common chemicals are ammonia (from deliming materials) and the nitrogen contained in proteinaceous materials (from liming/unhairing operations).

These sources of nitrogen pose two direct problems.

- Plants require nitrogen in order to grow, but the high levels released by substances containing nitrogen over-stimulate growth. Water-based plants and algae grow too rapidly, whereupon waterways become clogged and flows are impaired. As the plants die, a disproportionately high amount of organic matter has to be broken down. If the load outstrips the natural supply of oxygen from the river, plants, fish and aerobic bacteria die and ultimately anaerobic conditions develop.
- The nitrogen released through protein breakdown and the deliming process is in the form of ammonia. The latter can be converted by bacteria over several stages into water and nitrogen gas which is ultimately released into the atmosphere. Both of these breakdown products are non-toxic, yet large volumes of oxygen are needed in the process. If oxygen

demand is greater than the level supplied naturally by the water course, toxic anaerobic conditions can rapidly develop.

The nitrogenous compounds can be broken down by combining intensive aerobic and anoxic biological treatment. The oxygen demand is very high, thus leading to correspondingly high operational and energy costs. Calculations show that with typical tannery effluent, some 40% of the oxygen requirements are spent on removing the nitrogen component.

The compounds containing nitrogen can be determined by the Kjeldahl method of analysis. In this technique, all the nitrogenous matter is digested on being boiled with sulphuric acid to form ammonium compounds. In a second stage, the ammonia is released on being boiled with sodium hydroxide, and subsequently trapped in a boric acid solution. The level of ammonia released is determined by titration, and its value calculated as nitrogen.

vi. Sulphide (S^{2-})

The sulphide content in tannery effluent results from the use of sodium sulphide and sodium hydrosulphide, and the breakdown of hair in the unhairing process.

The sulphides pose many problems:

Under alkaline conditions, sulphides remain largely in solution. When the pH of the effluent drops below 9.5, hydrogen sulphide evolves from the effluent: the lower the pH, the higher the rate of evolution. Characterized by a smell of rotten eggs, a severe odor problem occurs.

Comparable in toxicity to hydrogen cyanide, even a low level of exposure to the gas induces headaches and nausea, as well as possible damage to the eye. At higher levels, death can rapidly set in and countless deaths attributable to the build-up of sulphide in sewage systems have been recorded.

Hydrogen sulphide gas is also fairly soluble. When absorbed, weak acids can form and cause corrosion. This weakens metal roofing, girders and building supports. In sewers, major problems can arise as metal fittings, structural reinforcements and pipework corrode.

If discharged to surface water, even low concentrations pose toxicological hazards. Sulphides can be oxidised into non-toxic compounds by certain bacteria in rivers; however, this creates oxygen demand which, if excessive, can harm aquatic life.

Sulphides in effluent can be determined in several ways. One of the most accurate methods relies on the acidification of suitable quantities of effluent to generate hydrogen sulphide. This is flushed through the apparatus using nitrogen gas, whereafter it is trapped and converted into zinc sulphide. The amount of sulphide is determined by titration.

vii. Neutral salts

Two common types of salts are to be found in tannery effluent.

a. Sulphates (SO_4^{2-})

Sulphates are a component of tannery effluent, emanating from the use of sulphuric acid or products with a high (sodium) sulphate content. Many auxiliary chemicals contain sodium sulphate as a by-product of their manufacture. For example, chrome tanning powders contain high levels of sodium sulphate, as do many synthetic retanning agents.

An additional source is created by removing the sulphide component from effluent by aeration since the oxidation process creates a whole range of substances, including sodium sulphate. These sulphates can be precipitated by calcium-containing compounds to form calcium sulphate which has a low level of solubility. Problems arise with soluble sulphates, however, for two main reasons:

- Sulphates cannot be removed completely from a solution by chemical means. Under certain biological conditions, it is possible to remove the sulphate from a solution and bind the sulphur into microorganisms. Generally, however, the sulphate either remains as sulphate or is broken down by anaerobic bacteria to produce malodorous hydrogen sulphide. This process occurs very rapidly in effluent treatment plants, sewage systems and water courses, if effluents remain static.
This bacterial conversion to hydrogen sulphide in sewage systems results in the corrosion of metal parts, and unless sulphate-resistant concrete will gradually erode.
- If no breakdown occurs, the risk of increasing the total concentration of salts in the surface water and groundwater runs is incurred.

Sulphate analysis is performed by adding barium chloride solution to a sample of filtered effluent. The sulphates are precipitated as barium sulphate and filtration; drying and calculation can determine the sulphate level.

b. Chlorides (Cl^-)

Chloride is introduced into tannery effluents as sodium chloride usually on account of the large quantities of common salt used in hide and skin preservation or the pickling process. Being highly soluble and stable, they are unaffected by effluent treatment and nature, thus remaining as a burden on the environment. Considerable quantities of salt are produced by industry and levels can rapidly rise to the maximum level acceptable for drinking water. Increased salt content in groundwater, especially in areas of high industrial density, is now becoming a serious environmental hazard.

Chlorides inhibit the growth of plants, bacteria and fish in surface waters; high levels can lead to breakdowns in cell structure. If the water is used for irrigation purposes, surface salinity increases through evaporation and crop yields fall. When flushed from the soil by rain, chlorides re-enter the eco-system and may ultimately end up in the ground water. High salt contents are only acceptable if the effluents are discharged into tidal/marine environments.

The level of salt as chloride under acid conditions can be determined by titrating a known volume of effluent with a silver nitrate solution, using potassium chromate as an indicator. Under neutral or alkaline conditions, excess silver nitrate is added. This excess is then determined by retro-titration with potassium thiocyanate, using ferric alum as the indicator.

viii. Oils and grease

During leather manufacture, natural oils and grease are released from within the skin structure. If fat liquor exhaustion is poor, some fatty substances may be produced through inter-reaction when waste waters mingle.

Floating grease and fatty particles agglomerate to form 'mats' which then bind other materials, thus causing a potential blockage problem especially in effluent treatment systems. If the surface waters are contaminated with grease or thin layers of oil, oxygen transfer from the atmosphere is reduced. If these fatty substances emulgate, they create a very high oxygen demand on account of their bio-degradability.

The presence of oils and grease is determined by shaking the effluent sample with a suitable solvent and allowing the solvent to separate into a layer on top of the effluent. These solvent dissolves fatty matter, and a quantity can be drawn off and evaporated until dry. The residual grease can be weighed and calculated.

ix. pH value

Acceptable limits for the discharge of waste waters to both surface waters and sewers vary, ranging between from pH 5.5 to 10.0. Although stricter limits are often set, greater tolerance is shown towards higher pH since carbon dioxide from the atmosphere or from biological processes in healthy surface water systems tends to lower pH levels very effectively to neutral conditions. If the surface water pH shifts too far either way from the pH range of 6.5 - 7.5, sensitive fish and plant life are susceptible to loss.

Municipal and common treatment plants prefer discharges to be more alkaline as it reduces the corrosive effect on concrete. Metals tend to remain insoluble and more inert, and hydrogen sulphide evolution is minimized. When biological processes are included as part of the treatment, the pH is lowered to more neutral conditions by carbon dioxide so evolved.

x. Other metals

Other metals which might be discharged from tanneries and whose discharge may be subject to statutory limits include aluminium and zirconium.

Depending on the chemical species, these metals have differing toxicities that are also affected by the presence of other organic matter, complexing agents and the pH of the water. Aluminium, in particular, appears to inhibit the growth of green algae and crustaceans are sensitive to low concentrations. Cadmium, sometimes used in yellow pigments, is considered highly toxic. It is

accumulative and has a chronic effect on a wide range of organisms. If present in drinking water, it can induce brittleness in bones.

xi. Solvents

Solvents originate from degreasing and finishing operations. Solvents in effluents discharged to surface waters can form a microfilm on the water surface, thus inhibiting the uptake of oxygen. Solvents break down in a variety of ways; some inhibit bacterial activity and remain in the ecosystem for extended periods of time. Analysis is highly specialized.

The following table presents the quantity and composition of wastewater for every step of the tanning process. Per ton of hide a total of 35 m³ wastewater is produced.

Process step	Amount of water	pH	COD	N _{kj}	Cr
	(m ³ /ton)	-	Kg COD/m ³	Kg N/m ³	Kg Cr/m ³
Pre-tanning					
Soaking	4-6	6-9	30-40	1-1.5	-
Unhairing, liming	5-9	12-13	40-60	3-5	-
Fleshing	1-3	-	-	-	-
Deliming, bating	5-7	8.5-9	5-8	3.5-4	-
Tanning					
Chrome tanning	0.5-1	3.8-4	2-3	0.3-0.6	0.5-5
Neutralization	1-1.5	4.5-4.7	2.5-3	0.5-0.8	0-1.0
Dyeing, fatting	3-4.5	3.8-4.5	5-6	0.2-0.3	0-5.0
Finishing					
Drying	3-6				
Finishing	1-2				
Cleaning	5				

3. Air pollution

The following table gives the emissions into the air during the tanning process. Few figures are available about the amount of air pollution. An important part of the air pollution by leather tanneries is caused by the need for energy. RIVM (1992) estimated the need for the Dutch tanneries at: 439 kWh (electricity) per ton of raw hides and 108 m³ of gas per ton of raw hides. Gas is used for heating. Table 1 gives the emissions into the air as a result of gas-combustion. No figures are available about the emissions into the air as resulting from the use of electricity.

H₂S may be emitted into the air when the pH of the process water is less than 7. During the finishing-process volatile organic compounds are used.

Process step	Air pollutants
Unhairing/Liming	H ₂ S
Deliming/Bating	NH ₃
Finishing	Solvents, formaldehyde
Heating with gas	CO
	CO ₂
	NO ₂

Pollution load

Due to variations in raw material, process, chemicals, water consumption, etc., it is small wonder that figures about pollution load in the literature vary a lot and should be interpreted very cautiously. The tables below and the chart on overleaf may give a general idea, the usual reference being one tonne of wet-salted hides

Example of mass balance of chemicals in leather processing kg/ tonne of wet-salted hide			
Chemicals	Added in process	In/on leather & splits	Wasted
Chrome oxide, Cr_2O_3	25	12	13
Organic tannins	25	20	5
Fat liquors	22	17	5
Dyestuffs	5	4	1
Acids, bases, salts	191	-	191
Tensides	3	-	3
Enzymes	5	-	5
Finishing products	100	12	88
Total	452	72	380

Table 1. An example of mass balance in leather processing

Parameter	Unit	Average total pollution load	Typical limits, surface waters
BOD_5	mg O_2 /l	2,000	30-40
COD	mg O_2 /l	4,000	125-250
Suspended solids (SS)	mg/l	2,000	35-100
Cr^{3+}	mg Cr/l	150	1.5-2.0
S^{2-}	mg S/l	160	1.0-2.0
Total nitrogen (TKN)	mg N/l	160	100
Cl^-	mg Cl/l	5,000	Locally specific
SO_4^{2-}	mg SO_4 /l	1,400	Locally specific
Oil and grease	mg/l	130	Locally specific
TDS*	mg/l	10,000	Locally specific
pH		6-9	5.5-9.5

* The main constituents of TDS are chlorides (Cl^-) and sulphates (SO_4^{2-}) listed earlier.

Table 2. An example of average total pollution load – concentration in combined raw effluent, conventional process, water consumption: 45 m³/tonne

To avoid possible confusion arising due to differences in water consumption, it is practical to indicate the amount of pollutants generated per tonne of raw-hide input. While generally lower water consumption is very desirable (nowadays in well managed tanneries, it is below 30 m³/tonne), it obviously results in considerably higher concentrations of pollutants.

Parameter	Typical pollution load, conventional process, kg/tonne of wet-salted hide	A tannery with a daily input of 5 t of wet-salted hides, conventional process, discharges per day approx
BOD ₅	90	450 kg
COD	180	900 kg
Suspended solids, SS	90	450 kg
Cr ³⁺	7	35 kg
S ²⁻	7.5	37.5 kg
Total Nitrogen (TKN)	7.5	37.5 kg
Cl ⁻	225	1125 kg
SO ₄ ²⁻	63	315 kg
Oil and grease	6	30 kg
TDS*	450	2250 kg
pH	6-9	

* The main constituents of TDS are chlorides (Cl⁻) and sulphates (SO₄²⁻) listed earlier.

Table 3. An example of pollution load, conventional process